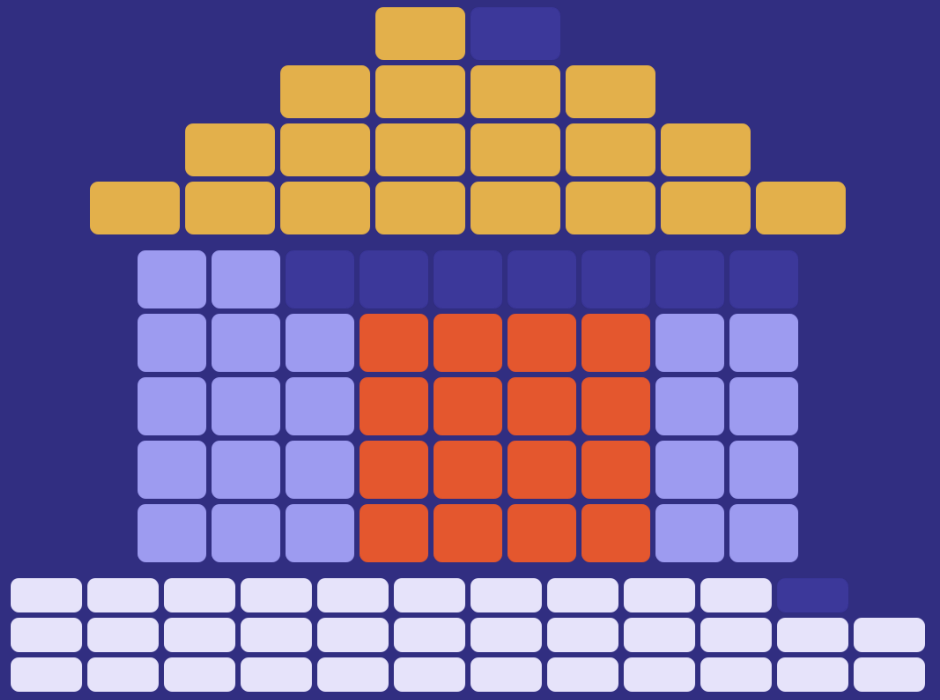
TrustKhaya

Your informal financial life, made visible.

Trust + ikhaya (home): trust that feels like home.



A real score: 91 of 100 bricks laid

FNB Hackathon submission | 26 October 2026

Mpho Dlamini · 067 602 9081

Trust you can build a home on

Trust

What a bank needs before it says yes. Young people already earn it in stokvels, wallets and shops.

+

Khaya

From ikhaya: home, in isiZulu and isiXhosa. A first account is a first foundation.

=

TrustKhaya

Trust that feels like home.



The mark: a house inside a ring of four arcs. The four arcs are the four trust signals around one home. The clay arc and the clay door are the guarantor who vouches.

Meet Thandi: no payslip, no bank account

Thandi is 22 and lives in Durban. She wants a first bank account, but she has no payslip, no utility bill in her name and no credit history.



Pays her stokvel

24 months, 23 of 24 payments made



Uses her wallet

Every month for 20 months, steady income



Repays the tuck shop

13 of 14 credit purchases repaid on time

A bank sees none of this, so Thandi looks like a stranger. TrustKhaya makes it readable.

Thandi is a fictional persona with simulated records, used to demonstrate the product.

Young South Africans are the gap

47.4%

of young people (15 to 34) looking for work cannot find it

Stats SA QLFS, Q2 2026

5 million

unemployed young South Africans: a payslip is the exception

Stats SA QLFS, Q2 2026

36.4%

of 15 to 24 year olds are not working, studying or training

Stats SA QLFS, Q2 2026

66.6%

of 15 to 24 year olds have an account, against 85.8% of those 25 and older

World Bank Global Findex 2025 (2024 data)

Overall, 81.1% of adults have an account, so almost 1 in 5 is still outside. The gap is widest for the young.

Sources: Stats SA Quarterly Labour Force Survey Q2 2026 (11 Aug 2026); NYDA statement on QLFS Q2 2026; World Bank Global Findex 2025.

The trust exists. Banks just cannot see it

Stokvels

R50 billion
moves through stokvels every year

810 000+
active stokvels

11 million+
members, about 40% of black adults

Spaza shops

150 000
spaza shops in South Africa

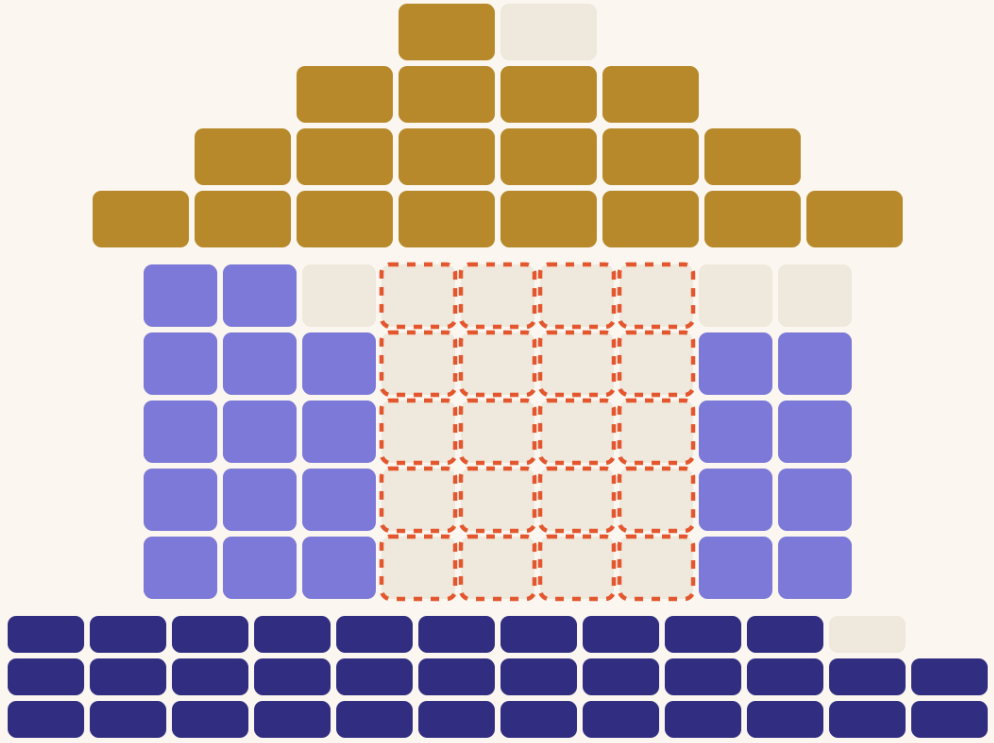
11.1 million
regular shoppers

R197 to R207 billion
informal retail market, mostly cash

A bank sees none of it. TrustKhaya makes it readable.

Sources: NASASA estimates (via Standard Bank, EWN); Trade Intelligence (via BusinessDay, 2024). Industry estimates, not audited.

Four signals, one house you can see



Thandi today: 75 bricks laid, the door still waiting

Foundation: stokvel — 35 points

How long, and how reliably, you pay into your stokvel or burial society

Walls: mobile wallet — 25 points

How long you have used it and how steady your income is

Roof: spaza credit — 20 points

How often you repay shop credit on time

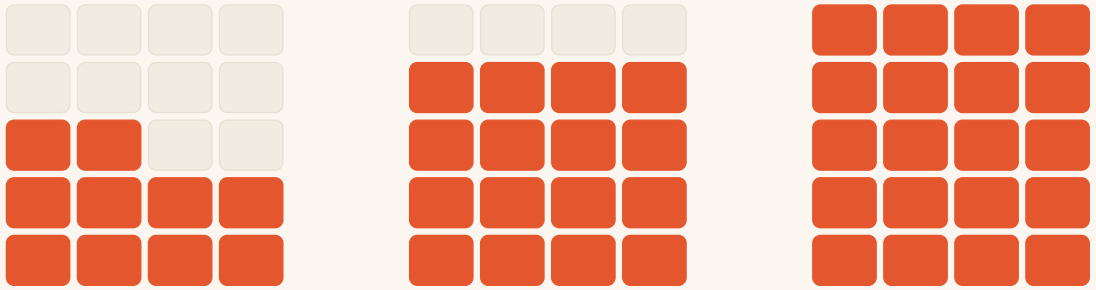
Door: guarantor — 20 points

A trusted elder backs you with a capped promise

Each point is one brick. The score is worked out from real records, never typed in by hand.

A capped promise, not just a referral

Gogo Ntombi backs Thandi. The most she could ever be asked to cover is the cap she chooses. Nothing more.



R500

10 of 20 bricks

R1 000

16 of 20 bricks

R2 000

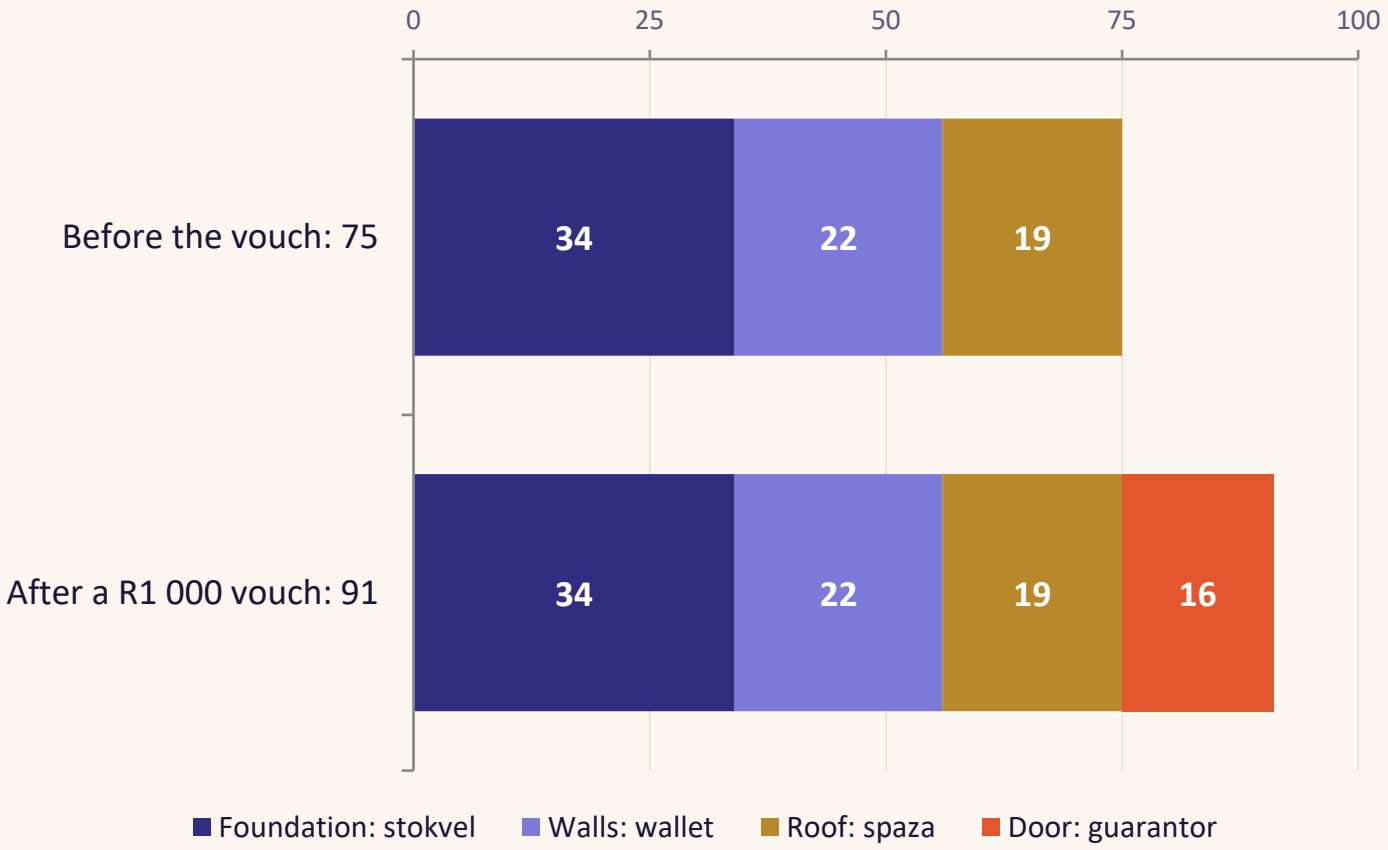
20 of 20 bricks

A bigger promise lights more of Thandi's door.

Safeguards, enforced on the server

- R5 000 total limit per guarantor
- Maximum of 3 active vouches
- No self-vouching: same person, phone or email is blocked
- Mutual vouching is flagged for the underwriter
- The guarantor sees a summary, never amounts

No black box: every point is explained



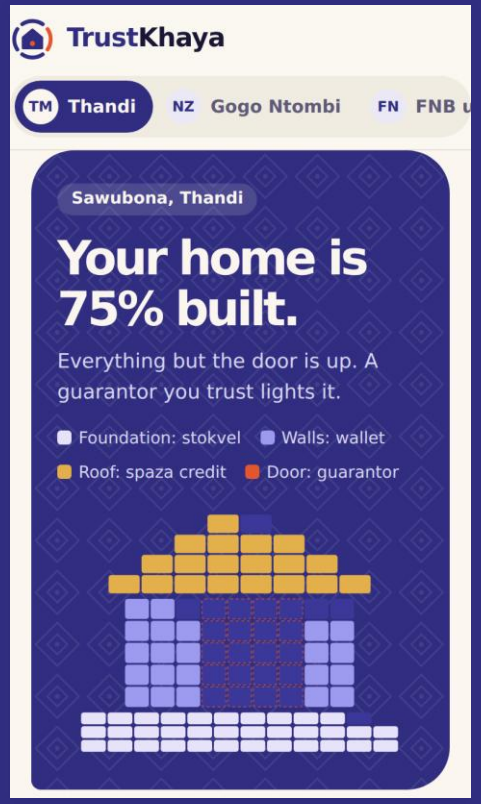
88% to 94%

Chance Thandi keeps her account in good standing, before and after the vouch. A model estimate on demo data.

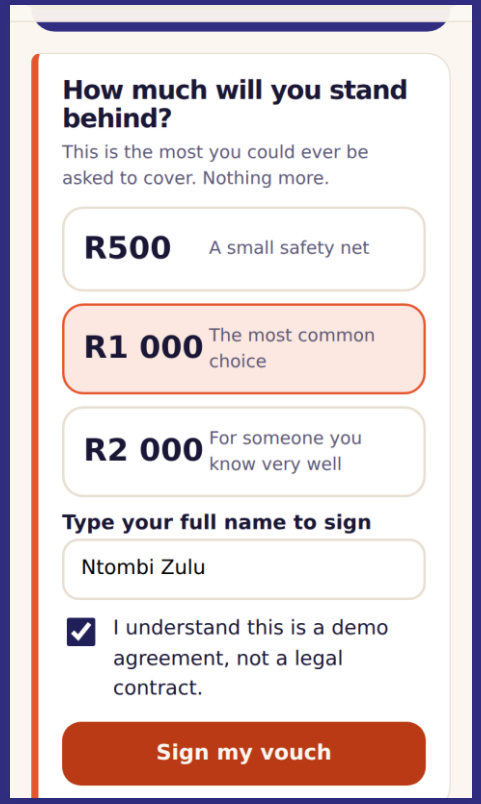
A human decides

The model explains. An FNB underwriter approves or declines, as POPIA s71 expects for automated decisions.

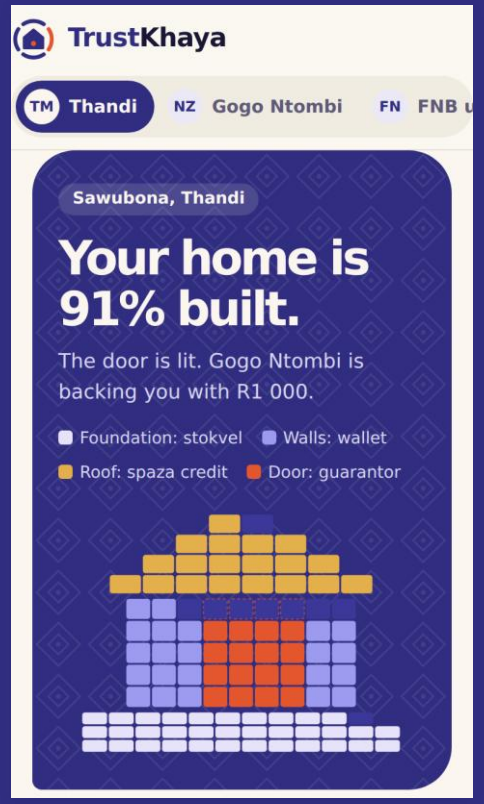
One story, three people, two minutes



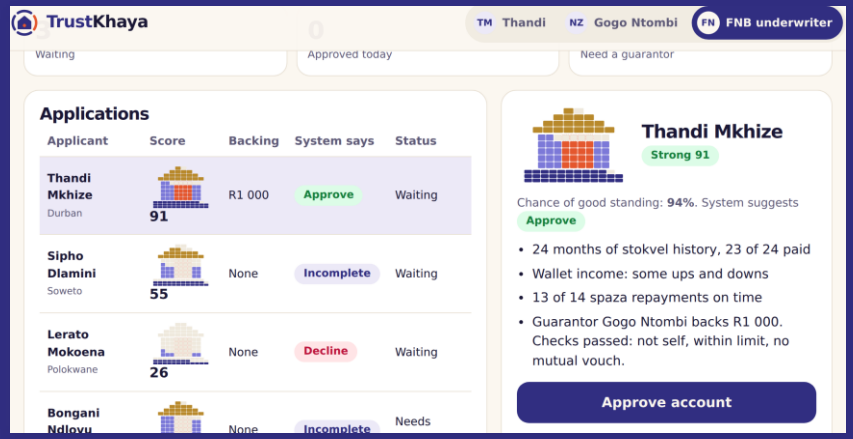
1. Thandi sees her home at 75%



2. Gogo Ntombi picks her cap



3. The door lights, Thandi is at 91%



4. FNB reads the reasoning and approves

Thandi gets a plain-language result: account approved, zero monthly fee.

Design reference screens of the product, with simulated demo data.

Technical architecture

ONE PURE ENGINE · ONE DATABASE SEAM · ONE HONEST UI

Browser · PWA
Next.js 14 App Router, React 19, TanStack Query, Tailwind, service worker

API routes
/api/* handlers · zod on every input · signed httpOnly session · persona authorisation

lib/repo.ts
The single database seam: every read and write, nothing else talks to storage

Supabase · Postgres
10 tables, RLS deny-all for the anon key, service role server-side only

lib/scoring.ts — pure, deterministic, no I/O
No clock, no randomness: the house, the explanation, the what-if preview and the underwriter snapshot cannot disagree. 164 tests pin it.

Verification on every change
Strict TypeScript, ESLint, production build, 164 Vitest tests (engine, safeguards, consent, portfolio, history, i18n), colour/contrast gate.

Demo runs on a seeded local adapter; the same schema and seed are applied to Supabase. No key or credential is exposed to the client.

Others prove alternative data works

TrustKhaya adds the missing layer: community liability, explained in plain words, underwritten by the bank.

Player	What they do	Capped community guarantor	Where TrustKhaya differs
Jumo	AI credit scoring and lending from mobile-wallet behaviour	No	Adds capped liability; the bank underwrites
Tala	Micro-loans scored from phone data	No	Uses community signals, not phone data
TymeBank	Zero-fee digital account, standard KYC	No	Solves who qualifies, not the fee
Capitec	Low-fee accounts; informal-market business credit	No	Targets the young individual, not the shop
FNB stokvel account	No-monthly-fee account for stokvel groups	No	Takes the same trust to each member
TrustKhaya	**Four signals + capped guarantor + explanation + bank decision**	**Yes**	**The combination is the product**

Public information, not an exhaustive review. Jumo and Tala repayment figures (above 90%) are company-reported. Sources: Jumo, Tala, Rateweb, Daily Maverick, FAnews.

FNB has the account. We unlock who can open it

New customers FNB cannot onboard today

The zero-fee account exists (FNB Easy Zero, R0 a month). The barrier is proof, not price.

Risk shared with the community

Each guarantor's exposure is capped, and the bank sees why every score is what it is.

A relationship from day one

A first account is the start of savings, cards and, later, credit.

The guarantee maths (illustrative)

100 accounts approved

90 stay in good standing (assumed)

10 fall behind

Each guarantee is capped at R1 000

At most R10 000 can be drawn on guarantees

No guarantor can stand behind more than R5 000 in total. Our revenue idea: a fee per approved account, priced below FNB's cost of acquiring that customer. The pilot sizes it with FNB's own numbers.

Illustration with an assumed 90% good-standing rate. Onboarding cost, revenue per account and default rates are bank-internal; the pilot measures them. FNB Easy Zero fee: Rateweb, 2026.

Built for the rules, honest about the gaps

The rules

- FICA is risk-based, so simplified onboarding of low-risk accounts is possible.
- POPIA s71 limits decisions made only by automation. A human underwriter decides and the reasons are explained.
- Consent is granular and can be withdrawn at any time.
- Needs legal review before launch: suretyship formalities (signed in writing) and National Credit Act treatment of guarantees.

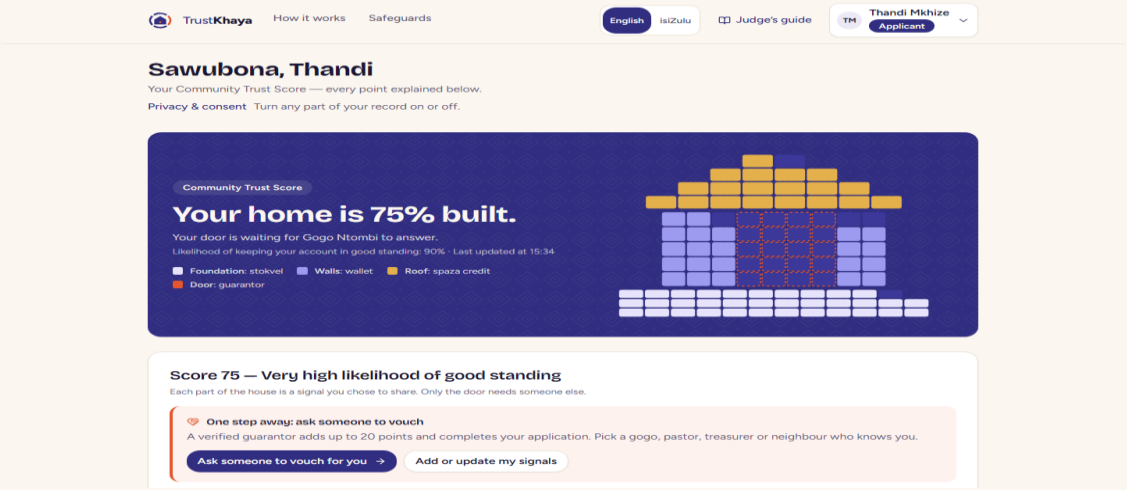
Security in the build

- Database key stays on the server, never in the browser.
- Every input validated; deny-by-default row-level security.
- httpOnly session cookies; an audit log on every change.
- Demo only today: PIN sign-in, simulated data, no penetration test.
- For production: FNB identity, dependency scanning, rate limiting, a threat model.

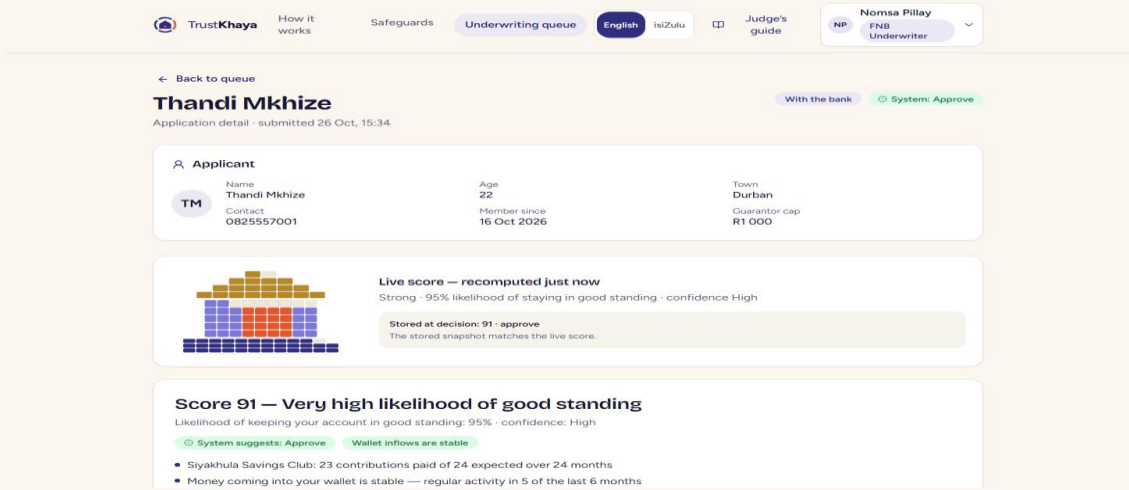
Sources: FIC Act 38 of 2001 and FIC guidance; POPIA section 71. Not legal advice.

The working app

FOUR SCREENS FROM THE LIVE PRODUCT

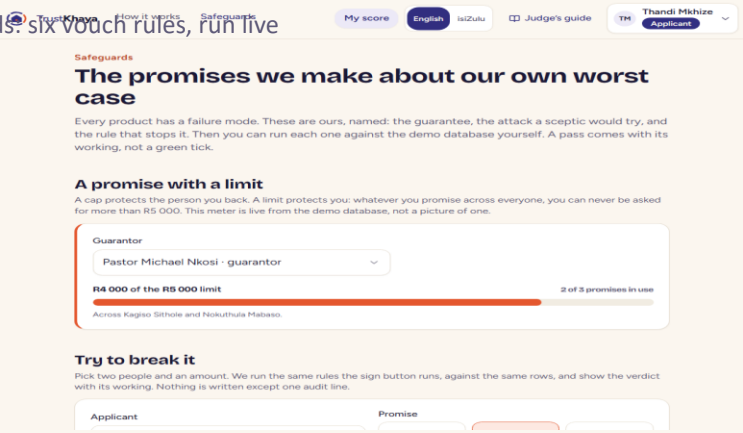


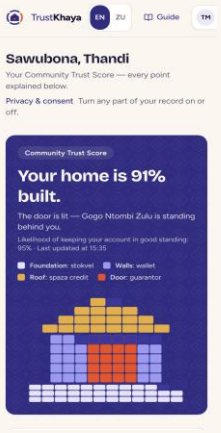
Applicant: 75 of 100, every brick attributed



FNB desk: score, delta, guarantor checks, decision

Safeguards: six vouch rules, run live





Mobile web app

Installable PWA: same product, same score, add to home screen. Live at graffgrid.co.za/work/trustkhaya.

From demo to pilot to FNB's front door

Today

Built and working

- Four signals and the house score
- Capped guarantor with safeguards
- FNB decision view and audit trail
- Installable mobile web app, isiZulu, WhatsApp share links

Next

Proposed 3-month pilot

- One stokvel partner and one FNB desk
- Real, consented data
- Legal review and fairness testing
- Measure approvals, 90-day good standing, guarantee draws

Later

Scale with partners

- Telco and open-finance data feeds
- WhatsApp Business and USSD for guarantors
- Onboard a whole stokvel at once
- More languages and portfolio analytics

Trust that feels like home.

The ask: a pilot with FNB and one stokvel partner.

Try it now

Live app

graffgrid.co.za/work/trustkhaya

Demo video (84 seconds)

Drive: drive.google.com/drive/folders/1xNLP09LDYrouG_6No26_f2HF0vn_y3fL

Try it in the browser — no install needed.

Code: github.com/Just4Skii/Mpho-Dlamini-Portfolio → projects/trustkhaya

TrustKhaya | Your informal financial life, made visible.
Mpho Dlamini · 067 602 9081